

Customer Support Ticket Priority Prediction

Problem Statement

Customer support teams struggle with managing large volumes of service requests due to manual ticket prioritization and assignment. This manual process causes delays in resolving urgent issues, inconsistent prioritization, and uneven workload distribution among agents.

Expected Outcome

- A centralized ticketing database to create, store, and manage customer support tickets.
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- AI-driven classification of tickets into High, Medium, or Low priority based on the issue description.
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- Automated routing and assignment of tickets to the appropriate support agents.
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- Automatic task generation for High-priority tickets to ensure immediate attention.
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- Role-based access controls to ensure secure access for agents and managers.
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- Manager dashboards and reports for monitoring ticket status, priority, workload, performance, and efficiency.
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Custom Object Configuration

- **Object Name:** Support Ticket Intelligence
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- **API Name:** `Support_Ticket_Intelligence__c`
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- **Fields:**
 - **Issue Description:** Text (`Issue_Description__c`)
 - **Priority Level:** Picklist (`Priority_Level__c`)
 - **Assigned Agent:** Lookup (`Assigned_Agent__c`)
- **Ticket Status:** Picklist (`Ticket_Status__c`)
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Auto-Launched Flow Logic (Modified for this Project)

1. Flow Type

Auto-Launched Flow

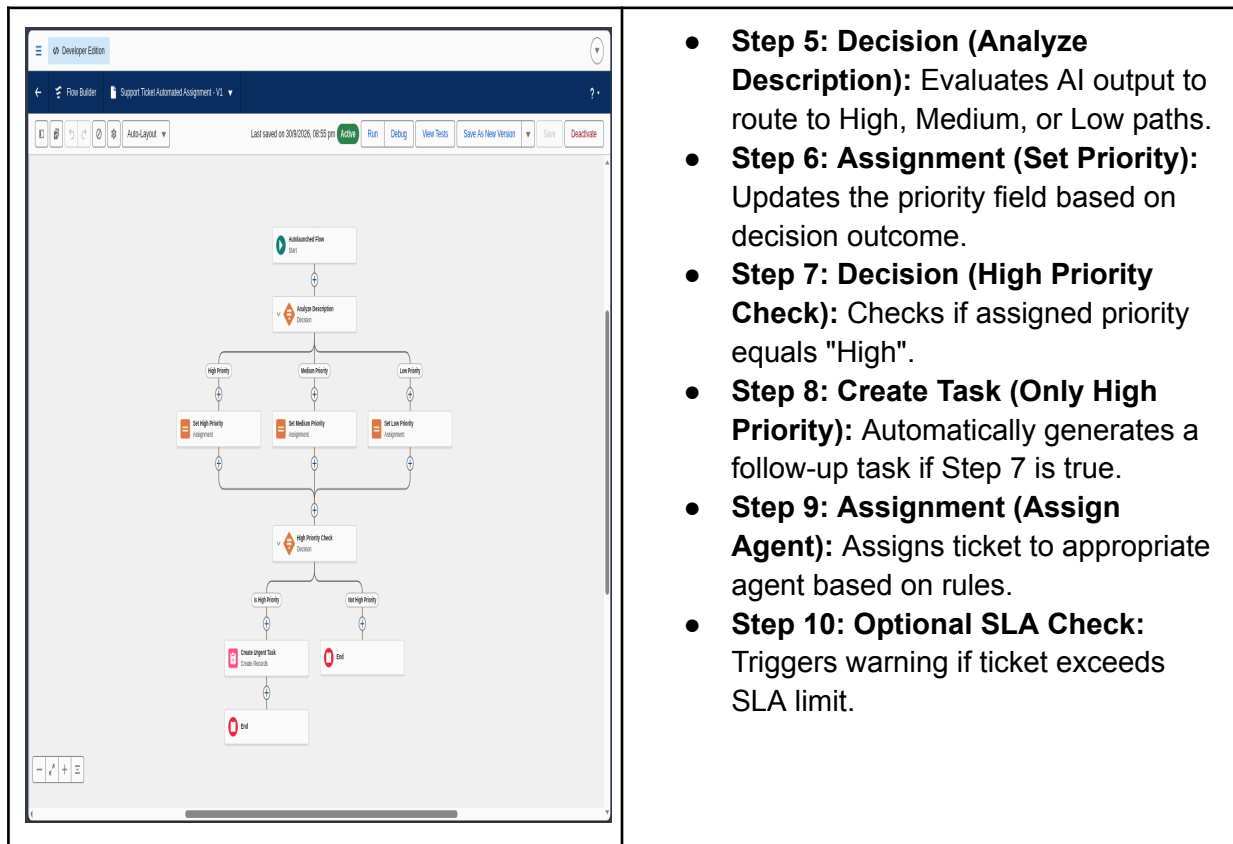
2. Variables

- **Input Variables:** `varAccountName`, `varTicketId`, `IssueDescription`
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- **Output Variables:** `varAccountId`, `varActionMessage`, `varAssignedTo`, `varPriorityLevel`
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3. Step-by-Step Flow Implementation

4. Final Flow Structure

The flow orchestrates the AI analysis, priority assignment, task creation, and final agent routing.



5. Final Outcome

The ticket is automatically updated with priority and an assigned agent, and urgent tasks are generated without manual triage.

Agentforce Configuration

- **Agentforce Subagent Configuration Name:** Priority Predictor Agent
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- **API Name:** Priority_Predictor_Agent
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- **Classification Description:** Analyzes customer support issue descriptions to predict urgency levels.
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- **Scope:** Reviews incoming ticket text to identify urgency keywords (e.g., "urgent", "not working", "failure").
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- **Instructions:** Analyze the provided 'Issue Description'. If the text indicates a critical failure or uses urgent terminology, return "High". If it indicates standard functionality issues, return "Medium". For general inquiries, return "Low".
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- **Agent Action Description:** Evaluates the issue text and returns the priority string to the Auto-Launched Flow.

Project Scope & Objectives

In-Scope (MVP):

- Creation of the `Support_Ticket_Intelligence__c` custom object to store tickets.
- Agentforce subagent configuration strictly for classifying issue descriptions into High, Medium, or Low priorities.
- Auto-Launched Flow to update the ticket priority, generate follow-up tasks for High-priority tickets, and assign tickets to a designated support queue.
- Basic manager dashboard/reports to view ticket volumes by priority level.

Out-of-Scope (Future Enhancements):

- Omnichannel routing (e.g., integrating live chat, phone, or social media tickets).
- Agentforce drafting automated email replies to customers (the AI is currently only classifying priority, not communicating externally).

Requirements & User Needs Analysis

- **Support Agents:** Need instant clarity on critical issues without reviewing every submitted description manually. They depend on automated workflows to create immediate follow-up tasks for urgent tickets and dispatch them directly to their assignment queue.
- **Support Managers:** Require high-level visibility across High, Medium, and Low priority distribution to balance team capacity and guarantee SLA compliance.
- **Customers:** Demand accelerated and consistent turnaround times, particularly when reporting critical outages or severe operational impact.

Core Salesforce Tools and Functionality

- **Custom Objects and Fields:** Design the `Support_Ticket_Intelligence__c` data structure along with necessary custom fields (such as Issue Description, Priority Level, Assigned Agent, and Ticket Status) to capture submitted service requests.
- **Agentforce (Einstein AI Builder):** Set up a specialized classification subagent (`Priority Predictor Agent`) that leverages large language model prompts to analyze incoming request details and assign an appropriate priority value.

- **Salesforce Flow (Auto-Launched):** Serves as the background automation process executing operational rules based on intelligence model outputs, including updating the priority level, creating follow-up task records, and reassigning ticket ownership.
- **Standard Task Object:** Utilized within automated workflows to instantly produce actionable items labeled "URGENT: Review Ticket" specifically for high-priority incidents.
- **Reports & Dashboards:** Provides leadership with operational metrics to observe incoming ticket trends, balance team assignments, and monitor SLA adherence.

Designing Data Model and Security Model

Data Model

- **Object Name:** Support Ticket Intelligence (`Support_Ticket_Intelligence__c`)
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- **Key Fields:**
 - **Issue Description (`Issue_Description__c`):** Captures incoming request details for model evaluation.
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 - **Priority Level (`Priority_Level__c`):** Stores predicted values (High, Medium, or Low).
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 - **Ticket Status (`Ticket_Status__c`):** Tracks progress across New, In Progress, and Resolved states.
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 - **Assigned Agent (`Assigned_Agent__c`):** Identifies the designated user handling the record.
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Security Model

- **Organization-Wide Defaults (OWD):** Configured to Private for custom ticket records to limit access strictly to assigned workload.
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- **Support Agent Access:** Granted Create, Read, and Update (CRU) permissions to generate new requests and maintain assigned items.
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- **Support Manager Access:** Configured with View All and Modify All permissions to monitor capacity, review analytics, and adjust ownership.
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- **Automation Permissions:** Provides background processes and intelligence components with context access to process issue details and assign priorities upon submission.